

Intelligent Transportation Systems (ITS) Analysis Modeling and Simulation (AMS) Community of Practice (CoP) – Summer 2026 Meeting

Wednesday August 26, 2026; 1:00 PM – 4:00 PM ET

Attendees (68):

USDOT (9):	Ian Berg (Volpe), James Colyar (FHWA), Anthony DePrator (FHWA), Thomas Fisher (FHWA-Michigan), John Hourdos (FHWA), Mark Parry (FHWA), Sarah Sun (FHWA), Kurt Walker (FHWA-Wisconsin), Jongsun Won (FHWA)
State/Local Agency (22):	Samuel Ahiamadi (Nevada DOT), Sana Arshad (Nevada DOT), W.D. Baldwin (Florida DOT), Chris Bell (Nevada DOT), Stan Brelan (Montana DOT), Juan Campos (Caltrans), Yang Chang (Wisconsin DOT), Emmanuel Posadas (Florida DOT), Cody Hamblin (Oklahoma DOT), Eric Harmer (Nevada DOT), Ron Hinson (South Carolina DOT), Mouyid Islam (Michigan DOT), Jason Levandoski (Alabama DOT), Chris Melson (Oregon DOT), Roxanne Mukai (Maryland Transportation Authority), Md. Mokhlesur Rahman (Baltimore Metropolitan Council), Guy Rousseau (Atlanta Regional Commission), Rajib Saha (Virginia DOT), Lisa Shemer (Maryland DOT), Rebekah Straub (Ohio DOT), Zifeng Wu (Caltrans), Nick Zinser (North Carolina DOT)
Academia (12):	Montasir Abbas (Virginia Tech), Marvin Baumann (Karlsruhe Institute of Technology), Mecit Cetin (Old Dominion University), Mohammed Hadi (Florida International University), Angshuman Guin (Georgia Tech), Kshitiz Khanal (NC State), Cami Li (University of Georgia), George List (NC State), Kaan Ozbay (New York University), Si Shi (NC State), Handong Yao (University of Georgia), Alexander Yoshizumi (NC State)
Consultant (13):	Joe Blasi (HNTB), Amir Ghiasi (Leidos), Ahmed Ghaly (Michael Baker International), David Hale (Leidos), Abdullah Kurkcu (Michael Baker International), Brad Miller (Michael Baker International), Keir Opie (Transpo Group), Raj Paradkar (Kimley Horn), Ossama Ramadan (WSP), Soheil Sajjadi (HNTB), Saeid Soleimaniamiri (Leidos), Zihang Wei (Leidos), Peng Zhu (KCI Technologies)
Vendor (2):	Dan Morgan (Caliper), Wuping Xin (Caliper)
Non-USDOT Federal Partner (1):	Erin Boyd (DOE)
Noblis Support (6):	Asean Davis, Rick Grahn, Claire Silverstein, Kathy Thompson, Peiwei Wang, Karl Wunderlich
Unknown (3):	“Ziyi Zhang”, “PL953th”, “SS903JD”

Agenda

Time (Eastern)	Topics	Speakers and Facilitators
01:00–01:10 p.m.	Welcome and Recap from Last Meeting	John Hourdos, FHWA
01:10–01:20 p.m.	Community Member Announcements	Members
01:20–01:50 p.m.	Introduction to Task Force Leads	Task Force Leads
01:50–02:20 p.m.	A High-Fidelity Microsimulation API for V2X Exploration	Wuping Xin, Caliper
02:20–02:25 p.m.	Break	
02:25–02:55 p.m.	AI for Transportation Systems Planning and Analysis	Alexander Yoshizumi, Kshitiz Khanal, Si Shi; NC State
02:55–03:15 p.m.	Traffic Analysis, Modeling, and Simulation: Impacts of Connectivity and Vehicle Automation	John Hourdos, FHWA
03:15–03:55 p.m.	Activity: How to Plan for CAVs?	John Hourdos, FHWA
03:55–04:00 p.m.	Wrap Up and Adjourn	John Hourdos, FHWA

Welcome and Recap from Last Meeting

John Hourdos (FHWA)

[John states purpose of community, provides recap from meeting held in April 2026, and presents overview of the GitHub forum located at https://github.com/FHWA/ITS_AMS_Community-of-Practice and asks CoP members to provide their email addresses to join the GitHub forum if they have not already done so]

Community Member Announcements

CoP Members

- **John Hourdos (FHWA):** FHWA is in the process of developing an AI Deployment Center of Excellence: <https://highways.dot.gov/ai-coe>.
- **Jongsun Won (FHWA):** The Traffic Analysis Roundtable Pool Fund study has started, and 14 State DOTs have joined this effort. If anyone is interested, reach out to leading agency Colorado DOT: <https://pooledfund.org/Details/Study/1810>.

- **Kaan Ozbay (NYU):** NYCDOT, FHWA, New York University's C2SMART Center, and the City College of New York's University Transportation Research Center are holding an *AI Applications in Urban Transportation Management Symposium* in September to convene industry professionals, public agencies, researchers, and roadway users:
<https://c2smart.engineering.nyu.edu/event/90899/>.

Introduction to Task Force Leads

Task Force Leads

Each task force provided an overview of task force focus/priorities.

- **Network/Operations Models (Research):** George List (NC State), Angshuman Guin (Georgia Tech)
 - Would like to publish a newsletter focused on data and data collection. Want to try to build support to do another data collection event, similar to the size of NGSIM.
- **Network/Operations Models (Practice):** Soheil Sajjadi (HNTB)
 - Vision: increase the flow of information between various practitioners (and between researchers and practitioners. More updates to come on how the Task Force will achieve this goal in the coming weeks.
- **AI/ML/Data Analytics:** David Hale (Leidos), Mohammed Hadi (FIU), and Zhitong Huang (VDOT)
 - The focus of the task force will be on four categories:
 - Publications – posting on forum/having discussion. Task force will prescreen.
 - Presentations – posting on forum/having discussion. Task force will prescreen.
 - Research opportunities – Announcements and funding opportunities posted on forum.
 - Road mapping – hold separate virtual meetings for discussion.
- **Safety:** Ossama Ramadan (WSP)
 - Provided a presentation with goals for the task force [[view presentation on forum](#)]. The group will look to address the safety analytics gap with a portfolio of safety focused AMS artifacts that FHWA can post on the forum to help practitioners make measurable better safety decisions. The three outlined principles are: credible methods, usable data, and practitioner uptake.
 - Planning for mixed vehicle fleets (connectivity, automation, human driver) is currently a hot topic at state DOTs.
 - Comment from Karl Wunderlich (Noblis): For people looking for an update on the ITS4US Program, please see: <https://www.its.dot.gov/research-areas/ITS4US/> There are four active sites in operational phase: NHTA (Buffalo), GDOT (Georgia), HIRTA (Iowa), and U-Washington (Seattle).

- Comment from Mouyid Islam (Michigan DOT): Regarding the mixed traffic environment, the vehicles with varying levels of automation interacting with vehicles not equipped with those, the drivers of vehicles of different levels of automation are "safer" (meaning self-selective in the vehicle fleet). Because these "safer" drivers select/ purchase these vehicles with automation for their safety. However, the ADS vehicles experience much fewer crashes than lower levels of automation.
- **Travel Demand / Scenario Planning:**
 - Montasir Abbas (Virginia Tech) expressed interest in leading this task force moving forward.

A High-Fidelity Microsimulation API for V2X Exploration

Wuping Xin (Caliper)

- Dr. Xin presented TransModeler's new V2X simulation API, designed to give researchers high-fidelity tools for exploring connected and autonomous vehicle scenarios [[view presentation on forum](#)]. The V2X-aware microsimulator exposes standard SAE J2735 concepts, including Basic Safety Messages (BSMs), Signal Phase and Timing (SPaT), MAP, and Roadside Unit (RSU) emulation, through three interface surfaces: a graphical UI for configuration and visual debugging, a COM-based API (TsmAPI) accessible from C#, Python, or C++, and a high-performance multithreaded C++ Vehicle Monitor for per-vehicle behavioral loops.
- Critically, the platform models real-world imperfections such as non-ideal RSU radio-frequency coverage patterns, packet loss, message staleness, GNSS position perturbation, pseudonym-based privacy strategies, and cybersecurity threats like message spoofing.
- A live demo showed a plugin built in .NET C# that visualized RSU coverage, tracked individual vehicles' signal timing windows, manipulated vehicle speed and lane-changing commands in real time, and displayed BSM generation rates and drop statistics.
- The API is slated for release in TransModeler 9, expected in beta within three to four months, and the demo plugin will be shared as open-source sample code.

AI for Transportation Systems Planning and Analysis State

Alex Yoshizumi, Kshitiz Khanal, Si Shi; NC

- A team from NC State's Institute for Transportation Research and Education (ITRE) presented on applying AI to transportation planning, spanning machine learning, natural language processing, computer vision, and large language models [[view presentation on forum](#)]. The primary case study presented was the Triangle Regional Model (TRMG2), a hybrid trip-based travel demand model for the Raleigh-Durham-Chapel Hill region (2+

million people), which uses decision trees in trip generation and boosting methods for non-home-based destination choice.

- Explainable AI (XAI) visualization tools were demonstrated through a suppressed-travel analysis comparing multiple model architectures. The team then presented an agentic documentation pipeline that chunks the TRM's 240,000-line codebase into manageable segments for an LLM (GPT-4o mini) to document iteratively — with self-evaluation, retry logic, and human escalation — producing a navigable website with side-by-side code verification.
- The team also highlighted broader applications including ML-based data imputation, LLM-assisted literature reviews, and visual language models for extracting unstructured information.
- **Question:** David Hale (Leidos): The 2 MPOs are Triangle West and Capital Area?
 - **Response:** Si Shi (NC STATE): Yes
- **Question:** Zifeng Lillian Wu (Caltrans): R square for Ordered Logit is the lowest?
 - **Response:** Yes
- **Question:** David Hale (Leidos): Does the TRMG2 use open-source ML libraries?
 - *Response not provided*
- **Question:** Mohammed Hadi (FIU): Were the models developed based on user surveys? Also, what is the time resolution? Is it the pea period?
 - **Response:** Yes, we use household travel surveys for the region (2-year increments).
- **Question:** Montasir Abbas (Virginia Tech): which software is this developed in?
 - **Response:** Alexander Yoshizumi (ITRE): TransCAD

Traffic AMS: Impacts of Connectivity and Vehicle Automation

John Hourdos (FHWA)

- The presentation outlined two megatrends 1) vehicle automation, which is advancing rapidly and nonlinearly, and 2) connectivity, which is lagging —and argued that both are reshaping macro-level choices (route selection, departure time, parking, destination) already influenced by navigation apps like Google Maps, as well as micro-level driving behaviors with long-term implications for housing, shopping, and travel patterns [[view presentation on forum](#)].
- John highlighted that navigation apps produce instantaneous, shortest path routing that game theory tells us cannot achieve system-optimal outcomes, and that AVs will amplify this by removing human judgment from the loop entirely. AVs also introduce new challenges: changed value of travel time (passengers can nap or work), zero-occupancy vehicle trips (cars roaming for parking), potential transit ridership loss, and incomplete network data leading to dangerous routing errors.
- The current vehicle fleet is overwhelmingly Level 2 automation with high automation near 0% and full automation limited to fleet-owned robotaxis in select cities. The Highway

Capacity Manual's Capacity Adjustment Factors for connected and automated vehicles require cooperative connectivity (vehicles accepting external instructions against their own logic) — a level of cooperation no manufacturer or user has agreed to yet. John posed this gap as the central question for the group: how should the AMS community plan for a 20-year horizon when the connectivity prerequisites for realizing capacity benefits remain unresolved?

- **Question:** Chris Melson (Oregon DOT): Are these two white papers published and available?
 - **Response:** Yes, they can be found on the forum:
https://github.com/FHWA/ITS_AMS_Community-of-Practice/discussions/19
- **Comment:** Karl Wunderlich (Noblis): If there are any comments and thoughts on the white paper, please place them in the discussion area on the GitHub forum.
- **Comment:** Kshitiz Khanal (ITRE): <https://www.nature.com/articles/s44284-026-00443-x>

Activity: How to Plan for CAVs? John Hourdos, Jongsun Won (FHWA); Karl Wunderlich (Noblis)

Participants were randomly assigned to three breakout rooms, each themed around a different planning horizon for CAVs:

- Near Term Planning (0-5 years): CAVs are happening now. What do you do in the next 2–3 years? How does it change your job today and tomorrow?
- Mid Term Planning (5-20 years): A project has been funded, consultants hired, and design/construction is underway. How do you evaluate CAVs when conducting a 20-year build/no-build analysis?
- Long Term Planning (30+): MPOs are planning 30+ years out, identifying projects for legislators to fund and place on 10-year construction lists. How do you account for CAVs at that level?

Near Term Planning (0-5 years) Breakout Room (John Hourdos, Asean Davis)

Participants: John Hourdos (FHWA), Asean Davis (Noblis), PL953TH, Anthony DePrator (FHWA), Angshuman Guin (Georgia Tech), Chris Melson (Oregon DOT), Joe Blasi (HNTB), Lisa Shemer (Maryland DOT), Mohammed Hadi (FIU), Nick Zinser (NCDOT), Samuel Ahiamadi (Nevada DOT), Soheil Sajjadi (HNTB), Wuping Zin (Caliper)

Question: What challenges do you see in the near term for CAV planning?

- Angshuman Guin: The problem for the near term is that the new AV vehicles are learning how to drive amongst human drivers. The systems are learning how the human drivers react and drawing on the interaction research that has been done at Georgia Tech.

- PL953TH: Want to be cautious of CAVs assumptions, because if they are not correct and are translated in to work products there will be some issues.
- John Hourdos: One challenge is we do not really know when the features are activated and when they are not activated. NHTSA knows when an event happens when the system is activated or not, but is this information communicated to safety engineers at state DOTs?
- Mohammed Hadi: In 2-3 years, we may also consider connected vehicles.
- Anthony DePrator: One challenge for higher level CAVs is what the vehicles could do in very unique situations at particular intersections and infrastructures. Suspect that as more advanced technology comes forward, we may see more capable higher-level vehicles.

Question: How do you plan for special events? Olympics? Super Bowl? World Cup? Planning is done a year or so in advance and how has this changed in the past 10 years.

- Anthony DePrator: If this is referring to changeable message signs, with CAVs it has changed a little, but this is more in emergency events and not so much special events. Having the capabilities to send out alerts to individual's phones.
- PL953TH: Found that most people do not change their emergency evacuation route based on new updates and alerts.

Mid Term Planning (5-20 years) Breakout Room (Jongsun Won, Kathy Thompson)

Participants: Jongsun Won (FHWA), Kathy Thompson (Noblis), Montasir Abbas (Virginia Tech), Marvin Baumann (KIT), Ian Berg (Volpe), Erin Boyd (DOE), David Hale (Leidos), Rebekah Straub (Ohio DOT), Sarah Sun (FHWA), Lilian Wu (Caltrans)

Question: What do you think will be the effect of CAVs on capacity/land use/maintenance?

- Rebekah Straub (Ohio DOT):
 - Capacity: Ohio did a study 5-6 years ago about the impact of CAVs on highway construction. Management asked if we can reduce the number of lanes with CAVs in the future. CAVs were added to disaggregate travel demand models and married with CDA and DTA models. Found that you needed an 85% penetration rate before a positive impact on capacity, otherwise it just added to congestion. In the end, they are still building the same number of lanes.
 - Land use: Ohio is a home-rule state. Each municipality determines what land use development happens; development happens more so in suburban areas with better school districts than necessarily in areas where CAVs are planned.

- David Hale (Leidos): Have a CDA regional simulation connecting micro and meso tools- looking to investigate this further and run tests on this to get data-driven estimations rather than macroscopic-driven estimations. Hope to have this info in the next year or two to get more concrete answers to a question like this.
- Ian Berg (Volpe): May not want to put CAV into one bucket as it can make the conversation more confusing. Many agencies think they do not need the connectivity piece. There are already other ITS solutions for CV application use cases like signal preemption, so analysis may not change much even if the CV component is removed.

Question: Are CAVs currently being considered in your initial design-level analysis?

- Rebekah Straub (Ohio DOT): we do not use CAVs in facility design in Ohio due to the uncertainty in CAVs. We use a K-30 Design Hour Factor in Ohio (FL uses a K-100 factor). We only have a couple research studies to show how CAVs will impact capacity or throughput or travel demand. Ohio is adding technology to the infrastructure, just not in the design hour of traffic volumes.
- Marvin Baumann (KIT): We are not considering CAVs so far in Germany, also due to the uncertainty. The German government did a recent project with VISSIM, and as they were preparing the final report, a new EU regulation came out saying that the CAVs had to behave like a human driver. Research projects are too slow to keep up with technological development.

Long Term Planning (30+) Breakout Room (Karl Wunderluch, Rick Grahn)

Participants: Karl Wunderlich (Noblis), Rick Grahn (Noblis), Ahmed Ghaly (Michael Baker International), Alexander Yoshizumi (ITRE), Amir Ghiasi (Leidos), Chris Bell (Nevada DOT), Thomas Fisher (FHWA), James Colyar (FHWA), Jason Levandoski (Alabama DOT), Mark Parry (FHWA), Md. Mokhlesur Rahman (Baltimore Metropolitan Council), Ossama Ramadan (WSP), Rajib Saha (Virginia DOT)

Question: Enter 3 words into the chat about what CV and AVs will look like in 2056?

- Ossama Ramadan (WSP): User in the loop – Vehicles will have higher levels of automation. Algorithms will take user input and preferences when determining route choice, preference to avoid traffic, scenic routes, stop every couple of hours for long trips, etc. User vs system optimum.
- Mark Parry (FHWA): Higher Density Freeways – shorter headways due to faster response times.
- Ahmed Ghaly (MBI): More accurate analysis and output
- Amir Ghiasi (Leidos): City sprawl

- Alexander Yoshizume (ITRE): Parking Facility Shifts – There is no reason to have so much parking in urban areas. Lots of opportunities for redevelopment. Parking locations will look very different based on private ownership or commercial fleet models. Also, parking could move between residential areas and downtown cores so that users/owners don't have to wait so long to be picked up.
- Jason Levandoski (ALDOT): Higher capacities and maintenance costs.
- Rajib Saha (VDOT): Land use in the modeling, and higher capacities
- James Colyar (FHWA): More efficient use of road space
- Chris Bell (NDOT): Increased capacity and more expansive land use
- Thomas Fisher (FHWA): Improved safety

Question: When do we need to start considering CVs and AVs in the project planning state?

- Ossama Ramadan (WSP): Now
- Mark Parry (FHWA): Now
- Alexander Yoshizumi (ITRE): Now, in that we should be revisiting our travel model architecture. We often try to pose questions that the models were not designed to answer.
- Rajib Saha (VDOT): Now
- James Colyar (FHWA): Now, but be wary of assumptions – some of the more optimistic assumptions may not be true. Easy to “consider” CAVs but need to be careful in how we do this.

Question: How critical is it to consider CVs and AVs when planning for 2050? (1 – not critical, 5 – critical)

- Ossama Ramadan (WSP): 5
- Mark Parry (FHWA): 3
- Amir Ghiasi (Leidos): 4
- Thomas Fisher (FHWA): 3
- Alexander Yoshizumi (ITRE): 4
- Jason Levandoski (ALDOT): 3
- Ahmed Ghaly (MBI): 3
- Rajib Saha (VDOT): 3
- James Colyar (FHWA): 3